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Studierichting: Informatica
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$$\int \arcsin x dx$$

Na partiele integratie:

$$= x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx$$

Na substitutie:

$$= x \arcsin x + \frac{1}{2} \int \frac{du}{\sqrt{u}}$$

$$= x \arcsin x + \sqrt{1-x^2} + C$$

$$\begin{aligned} \Rightarrow \int_0^1 \arcsin x dx &= \left[x \arcsin x + \sqrt{1-x^2} \right]_0^1 \\ &= \frac{\pi}{2} - 1 \end{aligned}$$

References